Ex.No.5 **SQUARE WAVE GENERATION USING IN-BUILT TIMER**

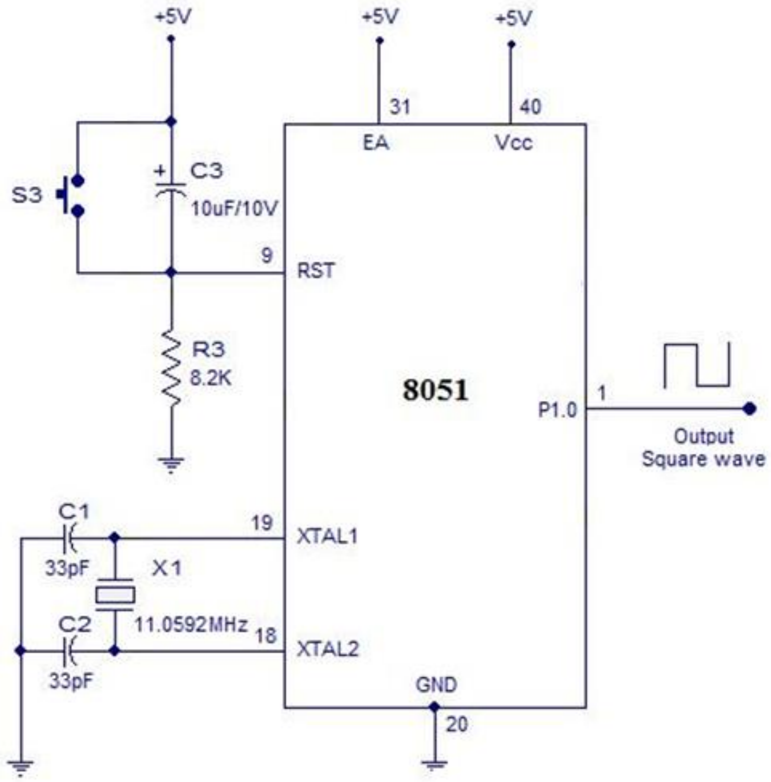
AIM

To generate square waveform using in-built timer of 8051 microcontroller

EQUIPMENT

Keil uvision-IDE, 8051 development board

CIRCUIT DIAGRAM



PROCEDURE

Step 1: Give a double click on µvision 5 icon.

Step 2: To create a new project, click on the project window and select new µvision project. Create a new folder with a project name to save the project.

Step 3: Select the target microcontroller i.e. 8051AH from Intel and click OK

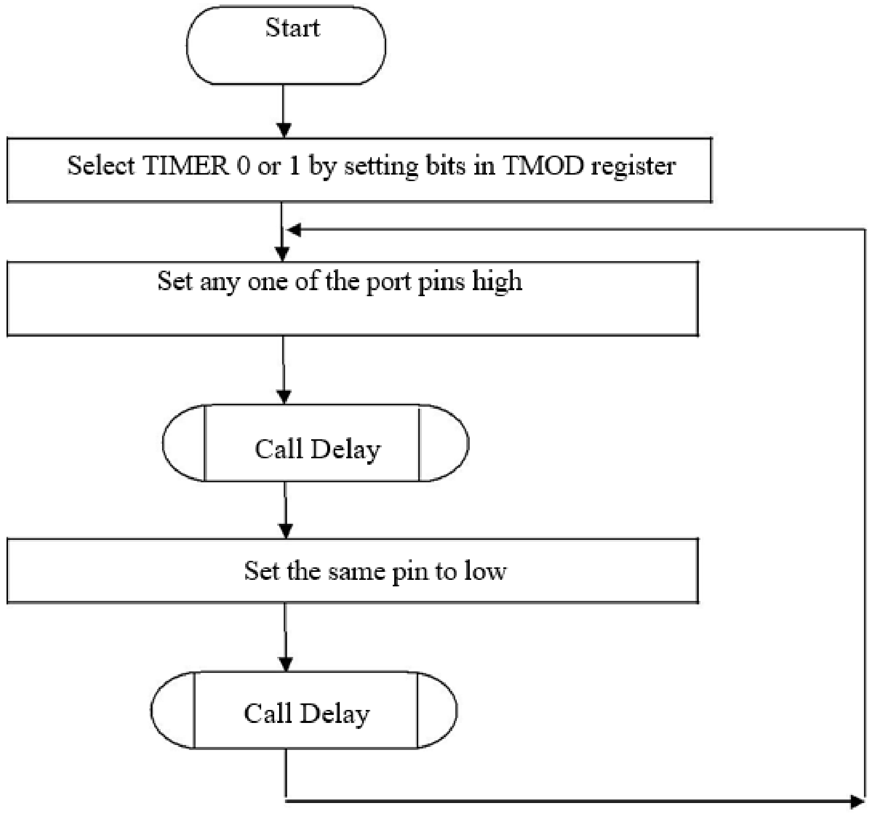
Step 4: Click YES to Copy STARTUP.A51 to Project Folder and Add file to Project. Type the Embedded C Program and save the file in the folder created with c extension.

Step 5: Add the .c File to source group by right click on Source Group, choose "**ADD files to Group**" option.

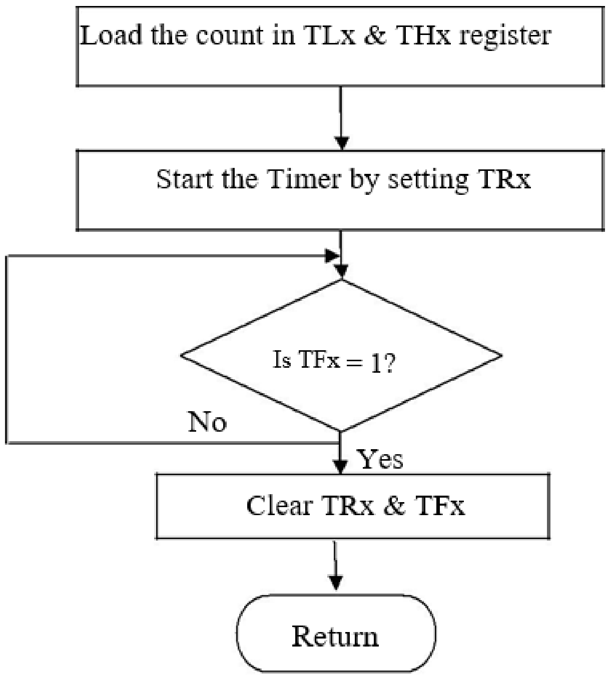
Step 9: To compile the project, click Project and select Build Target option or press F7.

Step 10: Click the logic analyzer and view the square waveform and measure the paramters.

FLOWCHART



DELAY ROUTINE USING TIMER



ALGORITHM

1. Start the program.
2. Initialize the header files.

3. Make any of the port pin as High.
4. Calculate the count value for delay
5. Select Timer 1 or Timer 0 and load the count value.
6. Make the selected port pin Low and again call the delay.
7. Execute the program
8. Verify the time delay of waveforms in the Logic Analyser with the calculated value and plot the graph.

DELAY CALCULATION

1) Divide the Time delay with Timer clock period.

Count = Time delay /1.085µs

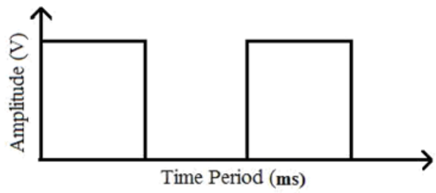
Timer clock period = 1 / (11.0592MHZ / 12)

2) Subtract the **Count** from 65536.

3) Convert the difference to **Hexadecimal** form.

4) Load this value into Timer register.

MODEL WAVEFORM:



PROGRAM

RESULT

Thus, the Square wave is generated using the Internal Timer of 8051 microcontroller.